Sri Eshwar
College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

Kondampatti (Post), Vadasithur (Via), Coimbatore - 641 202.

Department of

..

Record work of .... Laboratory

Certified bonafide Record of work done by

Name : .... Roll No. : ....

Class : .... Branch : ....

Reg. No. : ....

Place : *HOD* *Faculty In charge*

Date :

University Register No. ....

Submitted for the University Practical Examination held on

..

Internal Examiner *External Examiner*

Instructions for LABORATORY Classes

- Enter the Lab with observation, record notebook, calculator & necessary tools
- Enter the Lab with **CLOSED FOOTWEAR**
- Maintain Silence during the Lab Hours
- Boys should **"TUCK IN"** the Shirts
- Girls are instructed to wear **OVERCOATS**
- H**ANGING CHAINS, RINGS, WRIST WATCHES** and like items are likely to cause accidents and hence are to be **AVOIDED**
- **LONG HAIR** should be protected, let it not be loose especially near **ROTATING MACHINERY**
- Any machine / equipment **SHOULD NOT BE OPERATED** other than the prescribed one for the day
- Read and follow the work instructions inside the laboratory
- **POWER SUPPLY** to your table should be obtained only through the **LAB TECHNICIAN**
- Do not **LEAN** and do not be **CLOSE** to the rotating components
- Handle the equipments with care
- **TOOLS & GAUGE** sets are to be **RETURNED** before leaving the lab
- **ALL SUB - HEADING** and **DETAILS** should be neatly written on the right hand side page of the record
- For example

  Aim of the experiment

  Apparatus / Tools / Instruments required

  Procedure

  Model calculation

  Results / discussions / inference
- The sketches, block diagrams, circuits and flow charts are to be drawn on the left side blank page
- For example

  Neat Diagram

  Specifications / Design Details

  Tabulations

  Graph
- **GRAPHS / FIGURES** drawn on separate sheets in special cases, should be firmly pasted on to the record.
- Before writing the record, the student should get the corresponding observations signed by the **FACULTY - IN- CHARGE**
- The observation and record should be completed in all respects and submitted in the very **NEXT CLASS** itself.
- Begin every experiment on a **NEW PAGE**
- Write the **NAME OF EXPERIMENT** in Capital letters on the top of the page
- Experiment number with date should be written at the top left hand corner
- Be **PATIENT, STEADY, SYSTEMATIC** and **REGULAR**
- Follow the dress code for Laboratory classes
- Maintain punctuality

Record Index

S. No.	Date	Name of the Experiment	Page No.	Program (25)	Procedure (20)	Inference (20)	Viva (10)	Total (75)	Faculty Sign
							Average		

Signature of the Faculty member

VISION & MISSION OF THE DEPARTMENT

Vision:

"To emerge as a pioneering department, nurturing students into futuristic and globally competent Computer and Communication Engineering professionals."

Mission:

M1: To offer high-quality engineering education emphasizing practical outcomes.

M2: To foster innovative research and project development among students and instil a sense of social responsibility.

M3: To equip students with industry-relevant skills and robust leadership qualities through pedagogical methodologies and industry engagement.

M4: To inculcate a culture of socially relevant values and nurture moral, ethical and interpersonal skills among students.

Program Educational Objectives (PEOs):

PEO1: Acquire core competencies in computational technologies communication engineering with the ability to utilise the latest software tools to effectively address real-time challenges.

PEO2: Active involvement in research, innovation, or entrepreneurial ventures, resulting in significant societal contributions.

PEO3: Demonstrate industry-relevant skills, effective leadership qualities, and strong ethical values to exhibit professionalism and foster collaborative work experience.

Program Outcomes:

PROGRAM OUTCOMES (POs)		**Graduates of Computer and Communication Engineering willbe able to**
PO1.	**Engineering knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex software engineering problems.
PO2.	**Problem analysis**	Identify, formulate, research literature, and analyze complex software engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and software engineering sciences.
PO3.	**Design/development of solutions**	Design solutions for complex software engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4.	**Conduct investigations of complex problems**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5.	**Engineering tool usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex software engineering activities with an understanding of the limitations.
PO6.	**The Engineer and the world**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional software engineering practice.
PO7.	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of the software engineering practice.

PO8.	**Individual and Colloborative Teamwork**	Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9.	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO10.	**Project Management and Finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO11.	**Life-Long Learning**	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSO):

PSO1: Apply the concepts of computer and communication systems to develop software for different applications.

PSO2: Acquire skills and knowledge to solve real-time challenges in communication networks.

Course Outcomes COs:

CO. No.	Course Outcome	BTL	POs	PSOs
U23CC484.1	Interpret the wireless propagation models and implement these models to estimate path loss.	K3	1,2,3,4,5	2
U23CC484.2	Interpret cellular network concepts to enhance cellular network capacity.	K3	1,2,3,4,5	2
U23CC484.3	Compare mobile communication technologies, their multiple access mechanisms and simulate their performance differences.	K3	1,2,3,4,5,12	2
U23CC484.4	Illustrate mobile telecommunication system architectures, connection establishment and mobility management techniques.	K3	1,2,3,4,5	2
U23CC484.5	Demonstrate the key features of 5G, its architecture, and security mechanisms.	K3	1,2,3,4,5,12	2

CO & PO/PSO Mapping:

CO No.	Programme Outcomes												PSO	
	PO-01	PO-02	PO-03	PO-04	PO-05	PO-06	PO-07	PO-08	PO-09	PO-10	PO-11		PSO-01	PSO-02
U23CC484.1	3	2	1	1	2	-	-	-	-	-	-		-	2
U23CC484.2	3	2	1	1	2	-	-	-	-	-	-		-	2
U23CC484.3	3	2	1	1	2	-	-	-	-	-	1		-	2
U23CC484.4	3	2	1	1	2	-	-	-	-	-	-		-	2
U23CC484.5	3	2	1	1	2	-	-	-	-	-	1		-	1
Average	3	2	1	1	2	-	-	-	-	-	1		-	2

"3"—High, "2"—Medium, "1"—Low, "-"—No Correlation

Syllabus:

U23CC484 – WIRELESS MOBILE COMMUNICATION

List of Experiments: (30 periods)

1. Outdoor Propagation (CO1)
   (i) Implement Okumura and Hata models to estimate the path loss in urban, suburban, and rural areas.
   (ii) Compare their outputs with varying base station heights and mobile station heights using plots.
2. Frequency Reuse Simulation in a Cellular Network (CO2)
   (i) Simulate a cellular layout with hexagonal cells using plotting functions.
   (ii) Assign different frequency groups and visualize how frequency reuse reduces interference.
3. Simulate and compare TDMA, FDMA and CDMA for Wireless Communication. (CO3)
4. GSM Call Setup Simulation: Implement a simple GSM call setup process using state transition representation. Also, model steps like channel request, authentication, and call acceptance. (CO4)
5. 5G mmWave Channel Modelling (CO5)
   (i) Model a millimeter-wave (mmWave) wireless channel.
   (ii) Analyze the path loss, multipath fading, and signal strength variation in a 5G scenario.

Ex.No : 01 **OUTDOOR PROPAGATION**

Date :

AIM:

(i) To Implement Okumura and Hata models to estimate the path loss in urban, suburban, and rural areas.

(ii) To Compare their outputs with varying base station heights and mobile station heights using plots.

APPARATUS REQUIRED:

Computer with MATLAB / OCTAVE

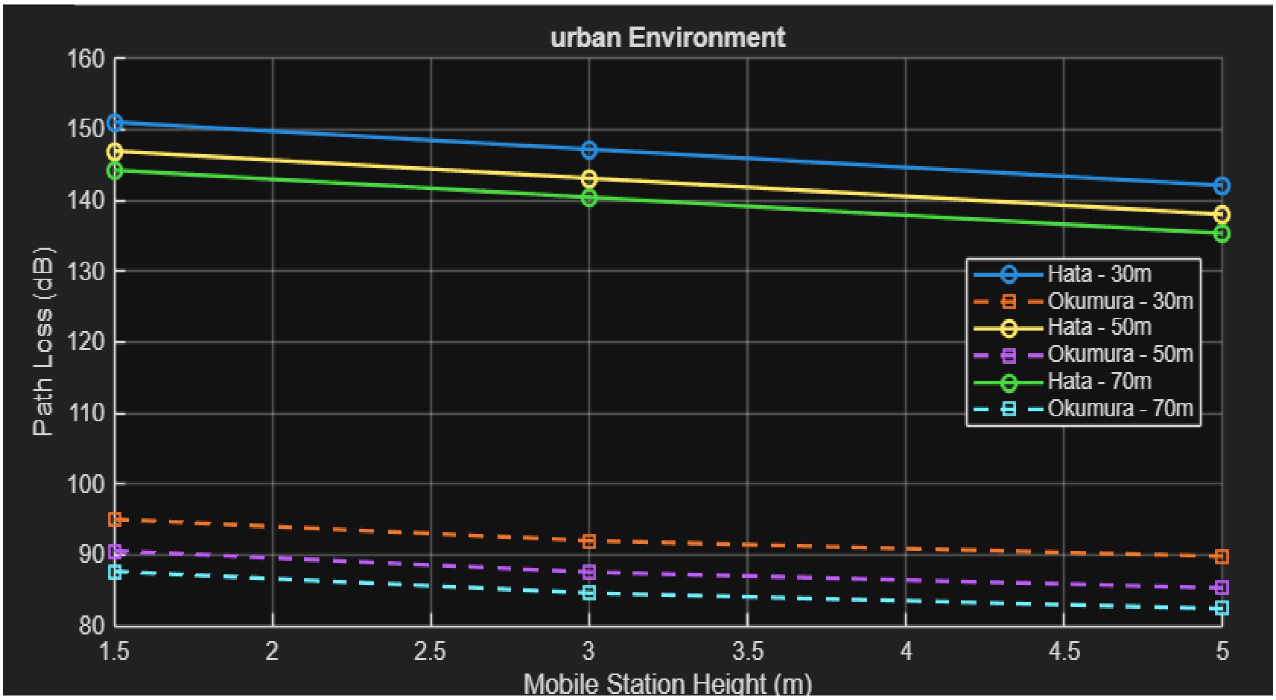
ALGORITHM:

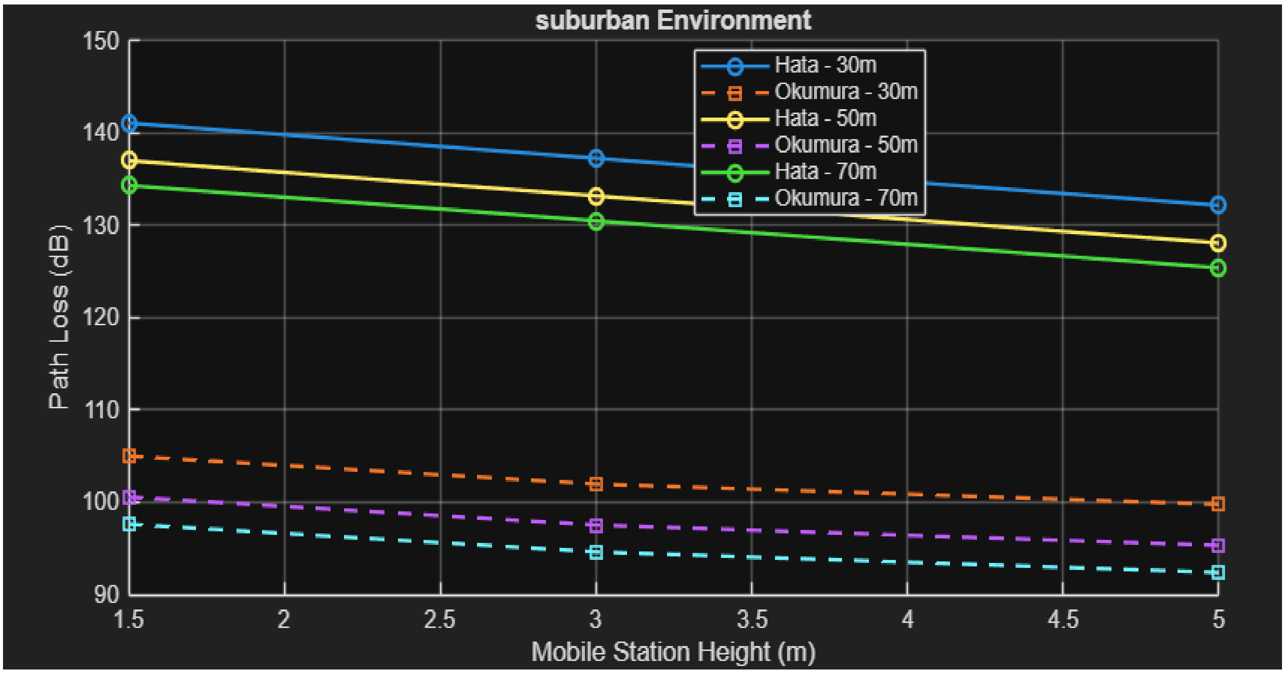
1. Start the program.
2. Input frequency, distance, base station height, and mobile station height.
3. Choose the area type: urban, suburban, or rural.
4. Calculate path loss using the Okumura model.
5. Calculate path loss using the Hata model.
6. Vary base and mobile station heights to analyze the effect.
7. Plot path loss for different heights and area types.
8. Compare the results of both models.
9. End the program.

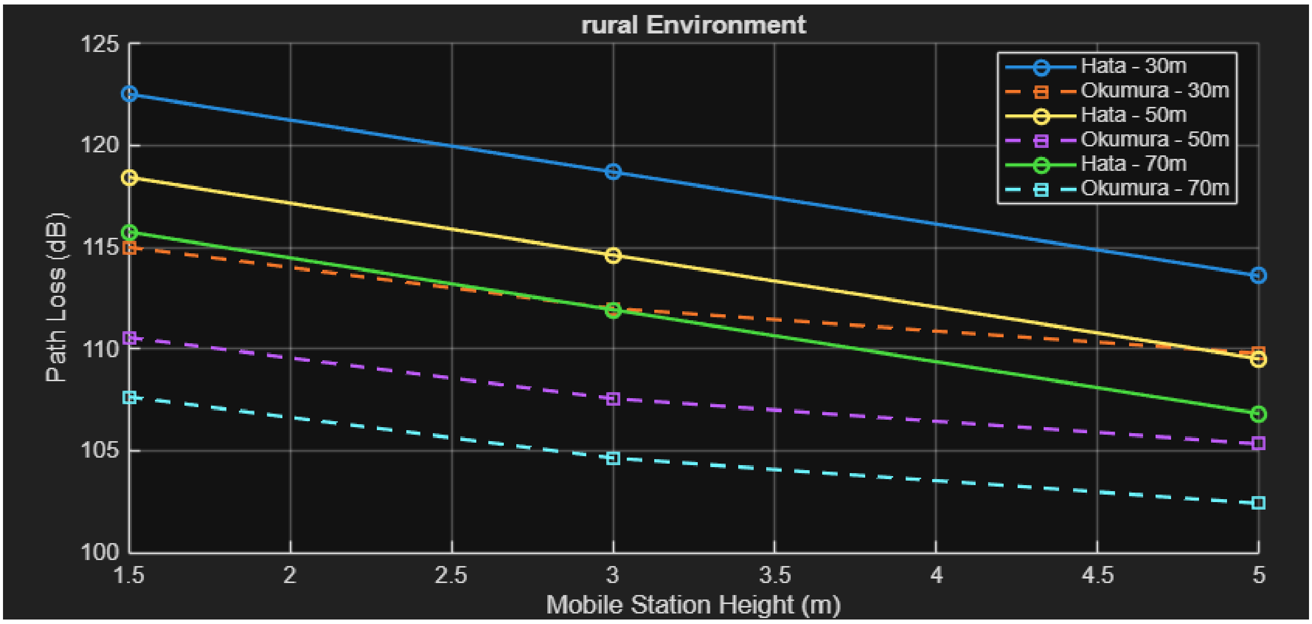
PROGRAM

```
clc;
clear;
close all;
f = 900;
d = 5;
env_types = {'urban', 'suburban', 'rural'};
hb_values = [30, 50, 70];
hm_values = [1.5, 3, 5];
Amu_map = struct('urban', 30, 'suburban', 20, 'rural', 10);
for k = 1:length(env_types)
    env = env_types{k};
    Amu = Amu_map.(env);
    figure;
    hold on;
    grid on;
    title([env ' Environment']);
    xlabel('Mobile Station Height (m)');
    ylabel('Path Loss (dB)');
    for i = 1:length(hb_values)
        hb = hb_values(i);
        hata = [];
        okumura = [];
        for hm = hm_values
            a_hm = (1.1 * log10(f) - 0.7) * hm - (1.56 * log10(f) - 0.8);
            L_hata = 69.55 + 26.16 * log10(f) ...
                    - 13.82 * log10(hb) ...
                    - a_hm ...
                    + (44.9 - 6.55 * log10(hb)) * log10(d);
            if strcmp(env, 'suburban')
                L_hata = L_hata - 2 * (log10(f/28))^2 - 5.4;
            elseif strcmp(env, 'rural')
                L_hata = L_hata - 4.78 * (log10(f))^2 ...
                        + 18.33 * log10(f) - 40.94;
            end
            Lf = 32.45 + 20 * log10(f) + 20 * log10(d);
            L_okumura = Lf ...
                        - 20 * log10(hb/200) ...
                        - 10 * log10(hm/3) ...
                        - Amu;
            hata = [hata, L_hata];
            okumura = [okumura, L_okumura];
            fprintf('%s | HB: %dm | HM: %.1fm -> Hata: %.2f dB | Okumura: %.2f dB\n', ...
                    env, hb, hm, L_hata, L_okumura);
        end
        plot(hm_values, hata, '-o', 'LineWidth', 1.5);
        plot(hm_values, okumura, '--s', 'LineWidth', 1.5);
    end
    legend({'Hata - 30m', 'Okumura - 30m', ...
        'Hata - 50m', 'Okumura - 50m', ...
        'Hata - 70m', 'Okumura - 70m'}, ...
        'Location', 'best');
end
```

OUTPUT


urban Environment
Path Loss (dB)
160
150
140
130
120
110
100
90
80
1.5
2
2.5
3
3.5
4
4.5
5
Mobile Station Height (m)
Hata - 30m
Okumura - 30m
Hata - 50m
Okumura - 50m
Hata - 70m
Okumura - 70m


suburban Environment
Path Loss (dB)
150
140
130
120
110
100
90
1.5
2
2.5
3
3.5
4
4.5
5
Mobile Station Height (m)
Hata - 30m
Okumura - 30m
Hata - 50m
Okumura - 50m
Hata - 70m
Okumura - 70m



Command Window

```
urban | HB: 30m | HM: 1.5m -> Hata: 151.02 dB | Okumura: 95.00 dB
urban | HB: 30m | HM: 3.0m -> Hata: 147.20 dB | Okumura: 91.99 dB
urban | HB: 30m | HM: 5.0m -> Hata: 142.10 dB | Okumura: 89.77 dB
urban | HB: 50m | HM: 1.5m -> Hata: 146.94 dB | Okumura: 90.57 dB
urban | HB: 50m | HM: 3.0m -> Hata: 143.12 dB | Okumura: 87.56 dB
urban | HB: 50m | HM: 5.0m -> Hata: 138.02 dB | Okumura: 85.34 dB
urban | HB: 70m | HM: 1.5m -> Hata: 144.25 dB | Okumura: 87.64 dB
urban | HB: 70m | HM: 3.0m -> Hata: 140.43 dB | Okumura: 84.63 dB
urban | HB: 70m | HM: 5.0m -> Hata: 135.33 dB | Okumura: 82.41 dB
suburban | HB: 30m | HM: 1.5m -> Hata: 141.08 dB | Okumura: 105.00 dB
suburban | HB: 30m | HM: 3.0m -> Hata: 137.26 dB | Okumura: 101.99 dB
suburban | HB: 30m | HM: 5.0m -> Hata: 132.16 dB | Okumura: 99.77 dB
suburban | HB: 50m | HM: 1.5m -> Hata: 137.00 dB | Okumura: 100.57 dB
suburban | HB: 50m | HM: 3.0m -> Hata: 133.18 dB | Okumura: 97.56 dB
suburban | HB: 50m | HM: 5.0m -> Hata: 128.08 dB | Okumura: 95.34 dB
suburban | HB: 70m | HM: 1.5m -> Hata: 134.31 dB | Okumura: 97.64 dB
suburban | HB: 70m | HM: 3.0m -> Hata: 130.49 dB | Okumura: 94.63 dB
suburban | HB: 70m | HM: 5.0m -> Hata: 125.39 dB | Okumura: 92.41 dB
rural | HB: 30m | HM: 1.5m -> Hata: 122.52 dB | Okumura: 115.00 dB
rural | HB: 30m | HM: 3.0m -> Hata: 118.69 dB | Okumura: 111.99 dB
rural | HB: 30m | HM: 5.0m -> Hata: 113.59 dB | Okumura: 109.77 dB
rural | HB: 50m | HM: 1.5m -> Hata: 118.44 dB | Okumura: 110.57 dB
rural | HB: 50m | HM: 3.0m -> Hata: 114.61 dB | Okumura: 107.56 dB
rural | HB: 50m | HM: 5.0m -> Hata: 109.51 dB | Okumura: 105.34 dB
rural | HB: 70m | HM: 1.5m -> Hata: 115.75 dB | Okumura: 107.64 dB
rural | HB: 70m | HM: 3.0m -> Hata: 111.92 dB | Okumura: 104.63 dB
rural | HB: 70m | HM: 5.0m -> Hata: 106.82 dB | Okumura: 102.41 dB
>>
```

Ex.No : 02 **FREQUENCY REUSE SIMULATION IN A CELLULAR NETWORK**

Date :

AIM:

- To simulate a cellular layout with hexagonal cells using plotting functions.
- To assign different frequency groups and visualize how frequency reuse reduces interference.

APPARATUS REQUIRED:

Computer with MATLAB / OCTAVE

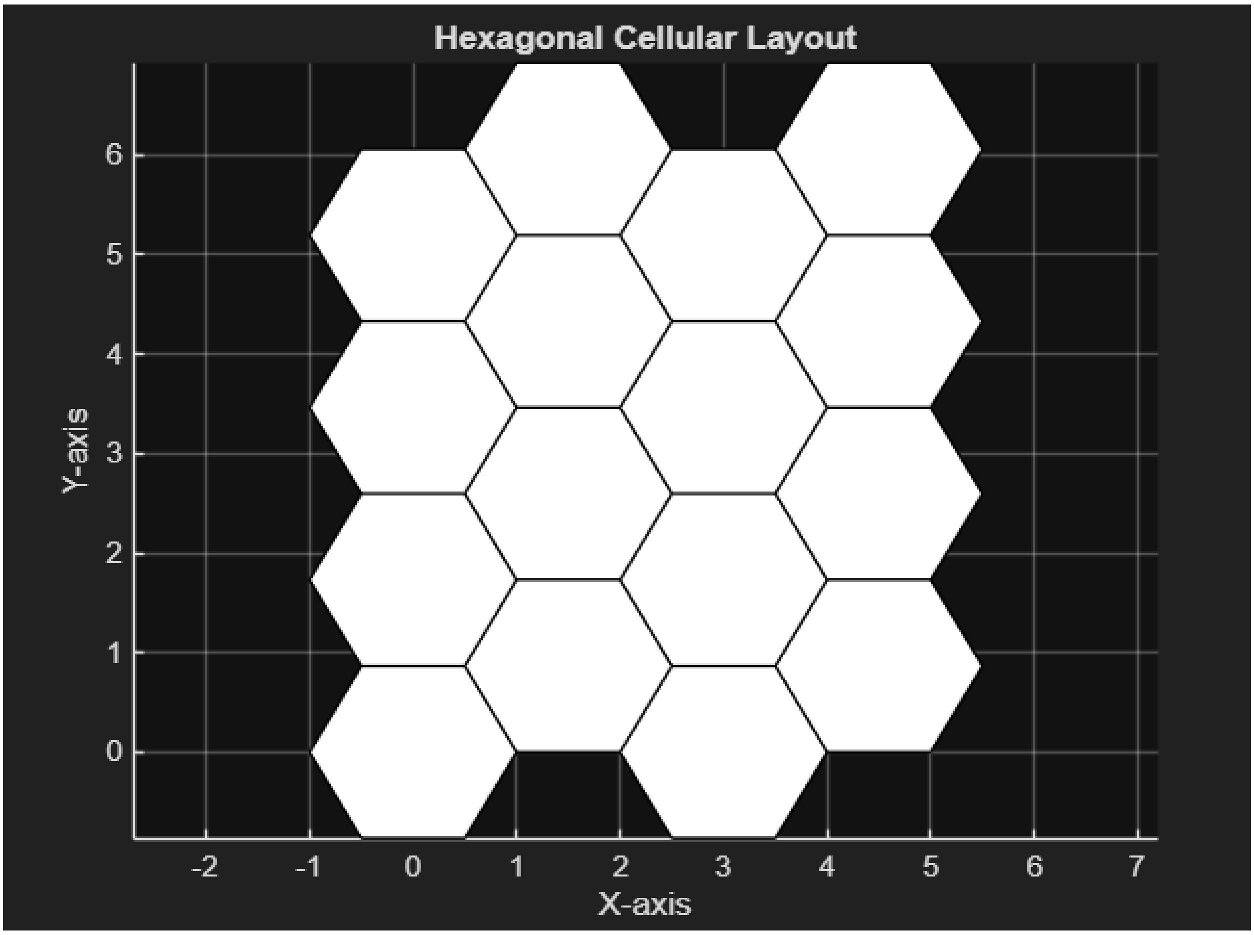
ALGORITHM:

1. Get the input sequence and its length.
2. Start the program.
3. Define the number of cells in a hexagonal layout.
4. Assign different frequency groups to cells using reuse pattern.
5. Plot the cellular layout using hexagons.
6. Color the cells based on assigned frequencies.
7. Visualize and identify frequency reuse.
8. Analyze how reuse reduces interference.
9. End the program.

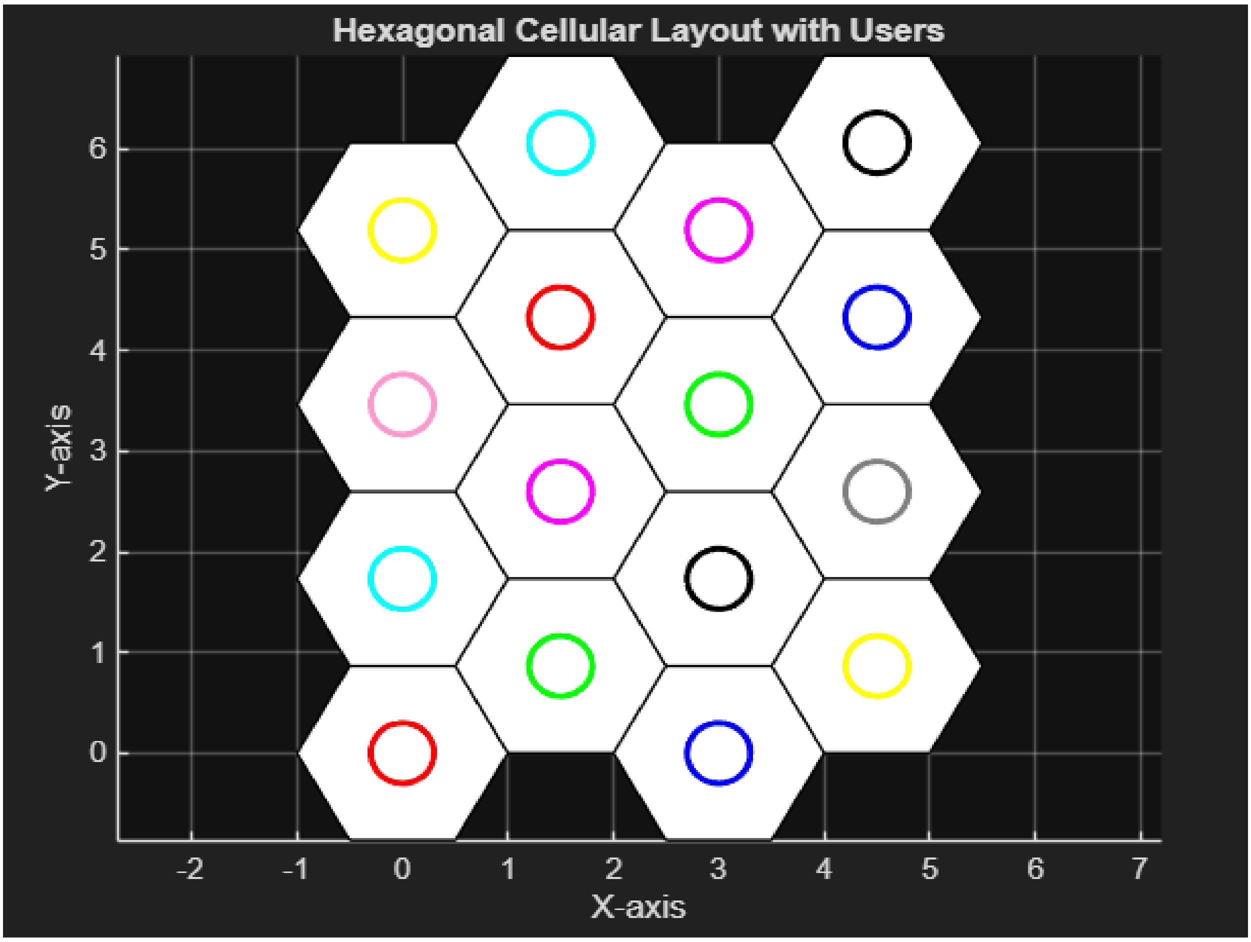
PROGRAM

```
clc;
clear;
close all;
R = 1;
N = 4;
theta = linspace(0, 2*pi, 7);
x_hex = R * cos(theta);
y_hex = R * sin(theta);
figure;
hold on;
for row = 0:N-1
    for col = 0:N-1
        x_off = col * 1.5 * R;
        y_off = row * sqrt(3) * R + mod(col,2)*(sqrt(3)/2 * R);
        fill(x_hex + x_off, y_hex + y_off, 'w', 'EdgeColor', 'k');
    end
end
axis equal;
grid on;
title('Hexagonal Cellular Layout');
xlabel('X-axis');
ylabel('Y-axis');
hold off;
figure;
hold on;

r_circle = 0.3;
theta_c = linspace(0, 2*pi, 100);
colors = {'r','g','b','y','c','m','k',[0.5 0.5 0.5],[1 0.6 0.8]};
count = 1;
for row = 0:N-1
    for col = 0:N-1
        x_off = col * 1.5 * R;
        y_off = row * sqrt(3) * R + mod(col,2)*(sqrt(3)/2 * R);
        fill(x_hex + x_off, y_hex + y_off, 'w', 'EdgeColor', 'k');
        x_circ = r_circle * cos(theta_c) + x_off;
        y_circ = r_circle * sin(theta_c) + y_off;
        color_idx = mod(count-1, length(colors)) + 1;
        plot(x_circ, y_circ, 'Color', colors{color_idx}, 'LineWidth', 2);
        count = count + 1;
    end
end
axis equal;
grid on;
title('Hexagonal Cellular Layout with Users');
xlabel('X-axis');
ylabel('Y-axis');
hold off;
```

OUTPUT



Hexagonal Cellular Layout
Y-axis
X-axis
6
5
4
3
2
1
0
-2
-1
0
1
2
3
4
5
6
7



Hexagonal Cellular Layout with Users
Y-axis
X-axis
6
5
4
3
2
1
0
-2
-1
0
1
2
3
4
5
6
7

Ex.No: 03 SIMULATION AND COMPARISION TDMA, FDMA AND CDMA FOR WIRELESS COMMUNICATION

Date:

AIM:

To Simulate and compare TDMA, FDMA and CDMA for Wireless Communication.

APPARATUS REQUIRED:

Computer with MATLAB / OCTAVE

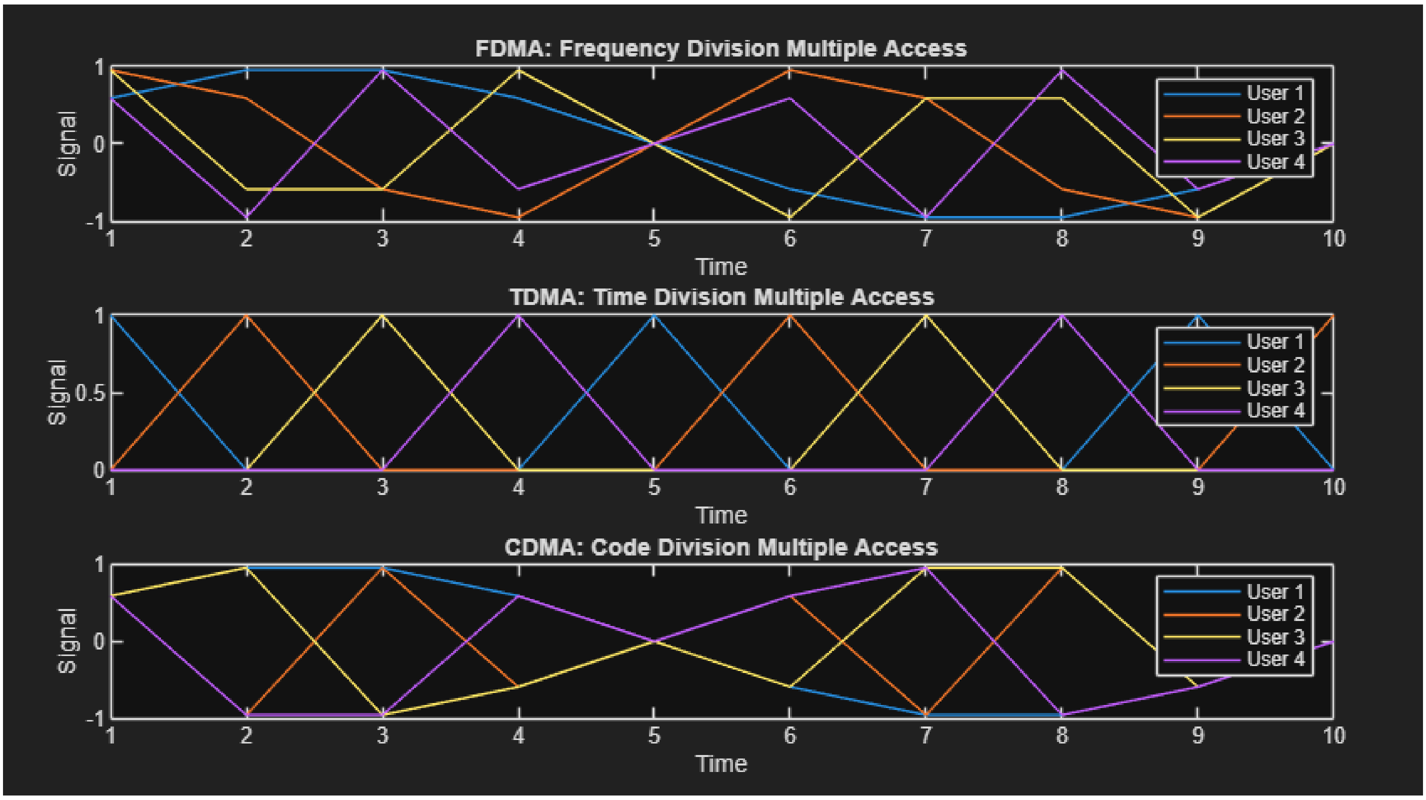
ALGORITHM:

1. Start the program.
2. Define users and total bandwidth.
3. Simulate TDMA by assigning time slots.
4. Simulate FDMA by assigning frequency bands.
5. Simulate CDMA using spreading codes.
6. Transmit signals for all three methods.
7. Plot the bandwidth usage and user access patterns.
8. Compare efficiency, interference, and capacity.
9. End the program.

PROGRAM

```
clc;
clear;
close all;
num_users = 4;
time_slots = 10;
fdma_signal = zeros(num_users, time_slots);
for user = 1:num_users
   fdma_signal(user, :) = sin(2*pi*user*(1:time_slots)/time_slots);
end
tdma_signal = zeros(num_users, time_slots);
for user = 1:num_users
   tdma_signal(user, :) = (mod(0:time_slots-1, num_users) == (user-1));
end
cdma_signal = zeros(num_users, time_slots);
codes = [
   1  1  1  1;
   1 -1  1 -1;
   1  1 -1 -1;
   1 -1 -1  1
];
for user = 1:num_users
   data = sin(2*pi*(1:time_slots)/time_slots);
   code = repmat(codes(user, :), 1, ceil(time_slots/4));
   code = code(1:time_slots);
   cdma_signal(user, :) = data .* code;
end
figure;
subplot(3,1,1);
plot(fdma_signal');
title('FDMA: Frequency Division Multiple Access');
xlabel('Time'); ylabel('Signal');
legend(arrayfun(@(x) sprintf('User %d', x), 1:num_users, 'UniformOutput', false));
subplot(3,1,2);
plot(tdma_signal');
title('TDMA: Time Division Multiple Access');
xlabel('Time'); ylabel('Signal');
legend(arrayfun(@(x) sprintf('User %d', x), 1:num_users, 'UniformOutput', false));
subplot(3,1,3);
plot(cdma_signal');
title('CDMA: Code Division Multiple Access');
xlabel('Time'); ylabel('Signal');
legend(arrayfun(@(x) sprintf('User %d', x), 1:num_users, 'UniformOutput', false));
```

OUTPUT



FDMA: Frequency Division Multiple Access
Signal
Time
User 1
User 2
User 3
User 4
TDMA: Time Division Multiple Access
Signal
Time
User 1
User 2
User 3
User 4
CDMA: Code Division Multiple Access
Signal
Time
User 1
User 2
User 3
User 4

Ex.No: 04 **GSM CALL SETUP SIMULATION**

Date:

AIM:

To Implement a simple GSM call setup process using state transition representation. Also, model steps like channel request, authentication, and call acceptance.

APPARATUS REQUIRED:

Computer with MATLAB / OCTAVE

ALGORITHM:

1. Start the simulation.
2. User initiates call request.
3. Base station sends channel request to MSC.
4. MSC performs authentication of the user.
5. On success, MSC allocates a traffic channel.
6. Call is connected between caller and receiver.
7. Show state transitions: idle → request → authentication → call setup → active.
8. End the call and return to idle state.
9. End the simulation.

PROGRAM:

```
clc;
clear;
close all;
states = {'Idle', 'Channel Request', 'Authentication', ...
         'Call Setup', 'Call Accepted'};
current_state = 'Idle';
for i = 1:5
   disp(['Current State: ', current_state]);
   current_state = transition(current_state);
end
disp(['Current State: ', current_state, ' - Call setup completed.']);
function next_state = transition(current_state)
   switch current_state
      case 'Idle'
         next_state = 'Channel Request';
      case 'Channel Request'
         next_state = 'Authentication';
      case 'Authentication'
         next_state = 'Call Setup';
      case 'Call Setup'
         next_state = 'Call Accepted';
      otherwise
         disp('Call setup process completed.');
         next_state = current_state;
   end
end
```

OUTPUT

```
Command Window
Current State: Idle
Current State: Channel Request
Current State: Authentication
Current State: Call Setup
Current State: Call Accepted
Call setup process completed.
Current State: Call Accepted - Call setup completed.
>>
```

Ex.No: 05 **5G mmWAVE CHANNEL MODELLING**

Date:

AIM:

- To model a millimeter-wave (mmWave) wireless channel
- To Analyze the path loss, multipath fading, and signal strength variation in a 5G scenario.

APPARATUS REQUIRED:

Computer with MATLAB / OCTAVE

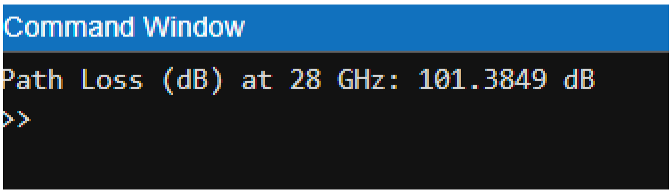
ALGORITHM:

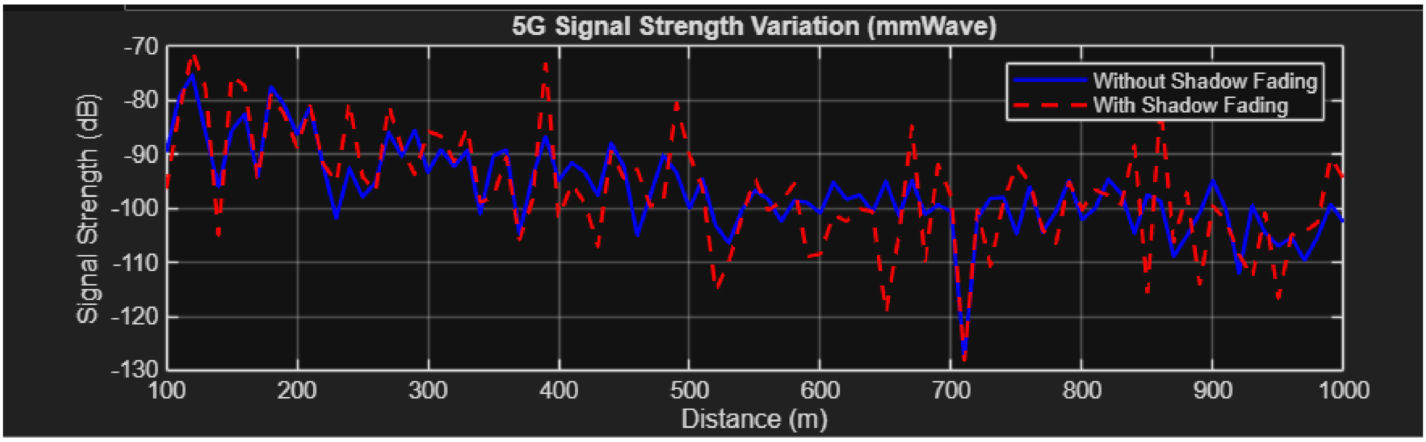
1. Start the program.
2. Define parameters: frequency (above 24 GHz), distance, environment type.
3. Model the mmWave channel using standard models.
4. Calculate path loss using mmWave equations.
5. Simulate multipath components and fading effects.
6. Analyze signal strength over varying distances.
7. Plot signal strength and fading characteristics.
8. Interpret results in a 5G environment.
9. End the program.

PROGRAM

```
clc;
clear;
close all;

%% ---------------- PART 1: FSPL (mmWave) ----------------

% Parameters
frequency = 28e9;          % 28 GHz
distance_fspl = 100;       % 100 meters
c = 3e8;                   % Speed of light
% Free Space Path Loss
path_loss = (4 * pi * distance_fspl * frequency) / c;
path_loss_dB = 20 * log10(path_loss);
% Display result
disp(['Path Loss (dB) at ', num2str(frequency/1e9), ...
      ' GHz: ', num2str(path_loss_dB), ' dB']);

%% ---------------- PART 2: 5G CHANNEL MODEL ----------------

% Parameters
distance = 100:10:1000;   % Distance range
n = 3.5;                  % Path loss exponent
shadow_std = 8;           % Shadow fading (dB)
SNR_dB = 20;
% Log-distance Path Loss Model
path_loss_dB = 20*log10(distance) + 20*log10(frequency) - 147.55;
% Rayleigh Fading
rayleigh = (randn(size(distance)) + 1i*randn(size(distance))) / sqrt(2);
fading_dB = 20*log10(abs(rayleigh));   % Convert to dB
% Signal strength in dB
signal_dB = SNR_dB - path_loss_dB + fading_dB;
% Shadow Fading (log-normal)
shadow_fading = shadow_std * randn(size(distance));
signal_with_shadow_dB = signal_dB + shadow_fading;

%% ---------------- PLOTTING ----------------

figure;
plot(distance, signal_dB, 'b', 'LineWidth', 1.5); hold on;
plot(distance, signal_with_shadow_dB, 'r--', 'LineWidth', 1.5);
title('5G Signal Strength Variation (mmWave)');
xlabel('Distance (m)');
ylabel('Signal Strength (dB)');
legend('Without Shadow Fading', 'With Shadow Fading');
grid on;
```

OUTPUT


Command Window
Path Loss (dB) at 28 GHz: 101.3849 dB
>>


5G Signal Strength Variation (mmWave)
Signal Strength (dB)
-70
-80
-90
-100
-110
-120
-130
100
200
300
400
500
600
700
800
900
1000
Distance (m)
Without Shadow Fading
With Shadow Fading